

Best Models Used for SHAP Analysis

Most frequently selected model by validation MCC across 10 seeds

Feature dataset	Most selected model	Selection frequency	Mean test MCC*
PCP + AAC	XGBoost	5 of 10 seeds	0.724
PCP + AAC + CTD	SVM-RBF	7 of 10 seeds	0.805
PCP + AAC + CTD + DPC	SVM-RBF	6 of 10 seeds	0.845

Overall SHAP choice: PCP + AAC + CTD + DPC with SVM-RBF

*Mean held-out test MCC of the validation-selected model in each seed; SHAP was calculated after model selection.

Other selected models: PCP + AAC – SVM-RBF (3), kNN (1), RF (1); CTD – LogReg (2), XGBoost (1); DPC – LogReg (2), RF (2).